

FEDERATED LEARNING FOR CODE GENERATION WITH HALLUCINATION-AWARE FINE TUNING

A PROJECT REPORT

Submitted by

ABHINAVH P

PRABHAKARA ARJUN R

BUVANES SRIVARDAN K

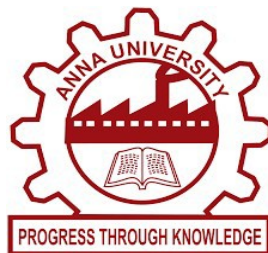
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BONAFIDE CERTIFICATE

Certified that the bonafide thesis report of the project work, **“FEDERATED LEARNING FOR CODE GENERATION WITH HALLUCINATION-AWARE FINE TUNING”** done by **Mr. Abhinavh P (2022503059), Mr. Prabhakara Arjun R(2022503003) & Mr. Buvanes Srivardan K (2022503037)** who carried out the project work under my supervision.

SIGNATURE OF SUPERVISOR

Dr. THANASEKHAR B

Professor

Department of Computer Technology

Anna University, MIT Campus

Chromepet, Chennai-600044

SIGNATURE OF HOD

Dr. PONSY R K SATHIA BHAMA

Head of Department

Department of Computer Technology

Anna University, MIT Campus

Chromepet, Chennai-600044

ABSTRACT

Large Language Models (LLMs) have demonstrated strong capabilities in automated code generation, enabling solutions for a wide range of software engineering tasks. However, these models often suffer from hallucinations, producing syntactically correct but semantically incorrect code, including invalid APIs and logical errors. Such issues reduce the reliability of LLM based systems in practical applications.

In this project, a federated learning based framework for improving code generation by reducing hallucinations in a privacy preserving manner is proposed. The system integrates hallucination detection to identify and correct errors in generated code, thereby creating high quality datasets for model adaptation. These datasets are used to fine tune the model using parameter-efficient Low Rank Adaptation (LoRA) across multiple decentralized clients.

Each client performs local fine tuning using its own data and only lightweight adapter parameters are shared with a central server, ensuring data privacy. Unlike traditional federated learning approaches that use standard Federated Averaging (FedAvg), this work introduces Federated Error Average (FedErrorAvg), where updates are weighted based on dataset size and hallucination error distribution across clients. This error-aware aggregation enables the model to learn more effectively from diverse error patterns, improving generalization and reducing hallucinations.

Experimental results demonstrate progressive improvement: total hallucinations reduce from 1,791 (baseline) to 1,635 (FedAvg) to 1,582 (FedErrorAvg), an 11.7% reduction. Pass rate improves from 52.4% to 61.2% to 62.6%. NameError drops 84%, SyntaxError 56% and TypeError 70% under FedErrorAvg, significantly outperforming FedAvg.

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ABHINAVH P (2022503059)

PRABHAKARA ARJUN R (2022503003)

BUVANES SRIVARDAN K (2022503037)

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
API	Application Programming Interface
APR	Automatic Program Repair
AST	Abstract Syntax Tree
BLEU	Bilingual Evaluation Understudy
CodeBLEU	Code Bilingual Evaluation Understudy
CSV	Comma Separated Values
DS-1000	Data Science 1000
FedAvg	Federated Averaging
FedErrorAvg	Federated Error Average
FedFT	Federated Fine Tuning
FL	Federated Learning
GPU	Graphics Processing Unit
IID	Independent and Identically Distributed
LLM	Large Language Model
LoRA	Low Rank Adaptation
MBPP	Mostly Basic Programming Problems
MLP	Multi-Layer Perceptron
NLP	Natural Language Processing
PEFT	Parameter Efficient Fine Tuning
RLHF	Reinforcement Learning from Human Feedback
SFT	Supervised Fine Tuning